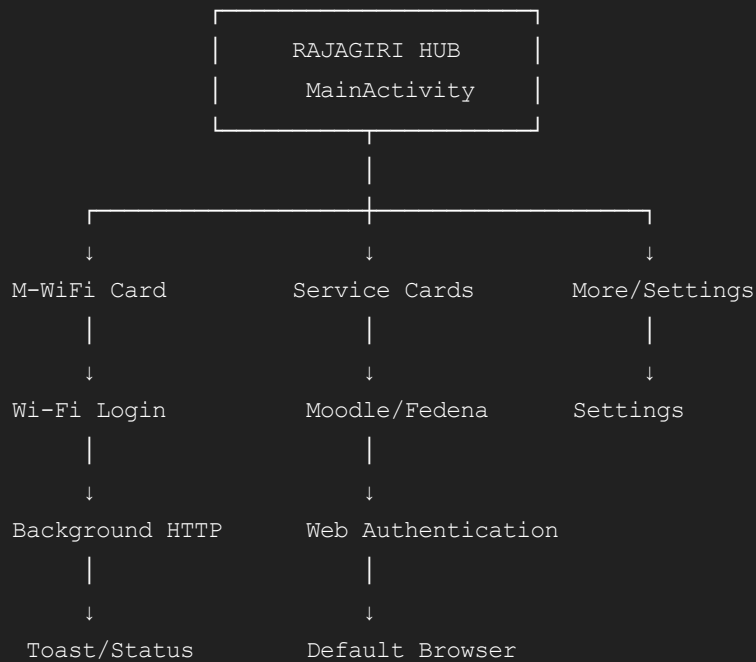


Yes. Before we continue the concept lessons, it's useful to have a **clear picture of what we're eventually building**. I wouldn't make it a huge app with 15 different screens—the strength should be a clean architecture around secure credentials and campus-service automation.

Planned app structure

I envision roughly this:

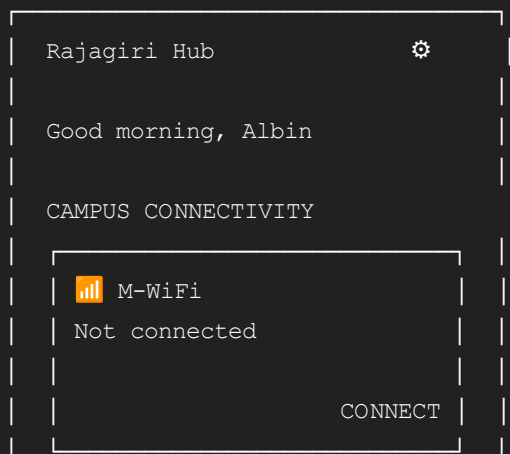


I'd plan **5 main areas/screens**.

1. 🏠 Home — Main Dashboard

This is the screen the user sees when opening the app.

Something like:



CAMPUS SERVICES

 Moodle LOGIN	 Fedena LOGIN
 Email OPEN	 Portal OPEN

Its job is simple:

M-WiFi

→ perform the automated authentication.

Moodle/Fedena

→ authenticate/open the service according to whatever authentication mechanism we establish.

Other services

→ simply open the appropriate website/app initially.

2. M-WiFi Credential / Setup Page

This is reached when the user first configures M-WiFi.

For example:

← M-WiFi


M-WiFi

Save your campus Wi-Fi
credentials securely.

Username

25204008

Password

.....

[SAVE]

The important thing is that **this page doesn't perform the login itself.**

It is responsible for:

Enter credentials
↓
Validate
↓
Encrypt
↓
Store in Room/SQLite

Then the Home screen can retrieve those credentials when the user presses **CONNECT**.

3. Credential Manager

I actually think we should have a **generic credential management page**, rather than creating separate pages for Moodle, Fedena, etc.

For example:

← Credentials

SAVED SERVICES

 M-WiFi
Username: 25204008
>

 Moodle
Username: albin...
>

 Fedena	
Username: albin...	
>	
+ Add Service	

Tapping a service:

Moodle
↓
Edit Credential

This is where the **encrypted database** becomes meaningful.

4. Service Detail / Credential Page

Rather than having separate hard-coded pages for each service, we can have one reusable page.

For example:

← Moodle
 Moodle
Website
moodle.rajagiri.edu
Username
albin...
Password
.....
[SAVE] [DELETE]

For M-WiFi, the page would show Wi-Fi-specific information instead.

This is where our **database CRUD concepts** will eventually be used:

```
CREATE
↓
new credential

READ
↓
retrieve credential

UPDATE
↓
change password

DELETE
↓
remove credential
```

5. 📖 Login History

This is optional for the basic version, but I **strongly recommend including it**.

It gives the database another meaningful purpose.

```
← Login History

TODAY

✓ M-WiFi
09:12 AM
Login successful

✓ Moodle
09:15 AM
Login successful

✗ Fedena
10:42 AM
Authentication failed

YESTERDAY

✓ M-WiFi
08:55 AM
```

Database:

```
LOGIN_HISTORY
-----
id
service_id
timestamp
status
message
```

This also lets you demonstrate database querying:

```
Show today's logins
Show failed attempts
Show Moodle history
Show latest login
```

6. Settings

A relatively small page:

```
← Settings

SECURITY

🔒 App Lock ON
👉 Biometric Unlock ON

CREDENTIALS

Manage Credentials >

HISTORY

Clear Login History >

ABOUT

Rajagiri Hub
Version 1.0
```

The biometric part would be something we add **after the basic application works**.

So how many Activities?

I wouldn't necessarily make every page a separate Activity.

A reasonable architecture could be:

```
MainActivity
|
├─ Home
├─ Credential Manager
├─ Service Detail
├─ History
└─ Settings
```

Depending on the architecture we choose, some of these can be Activities and/or Fragments.

We'll learn that distinction before implementing it.

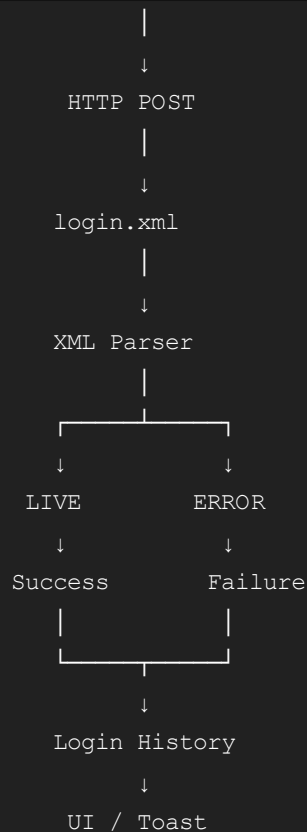
I don't want you blindly creating five Activities just because I drew five screens.

More importantly: the actual data flow

The UI is only the visible part.

The interesting architecture is underneath:

```
HOME
|
User clicks CONNECT
|
↓
Credential Repository
|
↓
Room Database
|
↓
Encrypted Credential
|
↓
Android Keystore
|
↓
Decrypted data
|
↓
M-WiFi Client
```



This is the real project.

The screens themselves aren't particularly difficult.

The technical value comes from what's happening behind them.

One change I'd make to your original idea

You originally said:

clicking the Wi-Fi thing can open a page where we enter credentials for first time

I'd make the experience slightly smarter.

First launch:

```
M-WiFi
  ↓
No credentials saved
  ↓
"Set up M-WiFi"
```

User enters credentials.

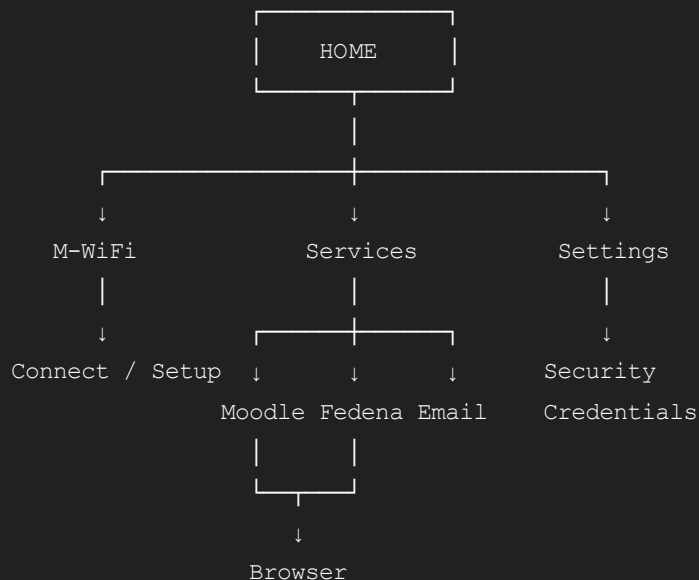
Subsequent launches:

```
M-WiFi
↓
Credentials already saved
↓
CONNECT
↓
Background authentication
↓
Toast
```

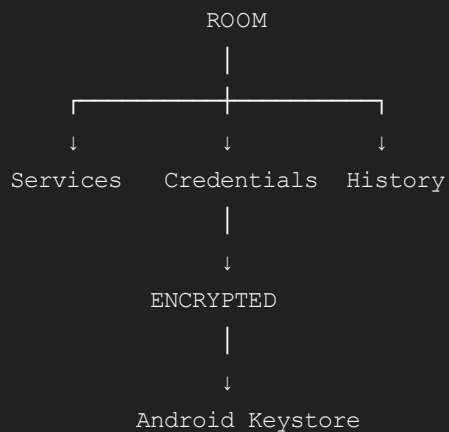
So the user doesn't repeatedly see the credential form.

Same principle for Moodle/Fedena.

The final navigation could be very simple



And the database sits underneath all of it:



One important caveat

For **M-WiFi**, we already have a concrete authentication mechanism from your PowerShell script.

For **Moodle/Fedena**, we'll **not design the final login mechanism yet**. Since you haven't tested their authentication flow, we'll investigate that when we reach web authentication. It could turn out that the best implementation is different from M-WiFi.

That actually makes the project more interesting: **one app, multiple authentication strategies**, rather than pretending every service works the same way.

For now, during the concept-learning phase, you don't need to build any of these screens. We'll continue with the Android fundamentals first.